Referential Regimes: Transformation-Based Identity Profiles for Neutral Substrates

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Abstract

Data systems operating under persistent legal, institutional, and analytic disagreement require shared reference without presupposing agreement about interpretation. This paper investigates identity preservation under transformation, building on the conditional neutrality result of *Neutral Substrates*.

We retain a finite catalog of candidate identity profiles and transformation families, while distinguishing stipulated classifications from model-derived results. Using invariant-defined identity bases and explicit effect models, we obtain six classification cells certified in Lean 4 across scope decomposition, rule refinement, and forking. The results demonstrate dependencies on field granularity, admissible transformation steps, and explicit continuity warrants.

We additionally characterize identity relations generated by invariant-preserving steps. Equality of these relations is equivalent to agreement on preservation of admissible steps in a shared comparison domain; a separating step provides a concrete witness of their difference.

The formalization makes identity claims auditable, provides testable conditions for systems transforming records, and exposes the assumptions responsible for each classification. Neither the finite catalog nor

the derived results establish necessary identity criteria, strict identity of intended referents, or a universal lower bound on identity regimes. Those conclusions require independent scientific grounding.

Keywords: formal ontology; identity profiles; transformation classification; individuation; co-reference; persistence; neutral substrates; referential common ground; stable reference

1 Introduction

Modern accountability systems increasingly operate under legal, institutional, and analytic disagreement. A shared record may need to support reference to the same intended subject while its representations are amended, forked, refined, decomposed, aggregated, or reclassified. Agreement about what a record means cannot be assumed, and neither can agreement about which identity conditions apply to its referents. This paper investigates candidate ways to characterize individuation, co-reference, and persistence under those transformations.

Neutral Substrates (Case 2026a) provides the motivating constraint: under Referential Common Ground and the stated foundational-layer restriction, a substrate satisfies semantic neutrality relative to its admissible frameworks. That conditional result does not establish a referential identity criterion for a particular bearer. The present paper examines what a declared criterion and selected transformation behavior can express, while keeping the adequacy of those choices open to independent justification.

The analysis pairs proposed identity bases with classifications of selected transformation families. An identity regime, in the terminology of this paper, is a declared basis together with its transformation profile. A classification records a proposed preservation, breakage, neutrality, or inapplicability judgment; it does not independently prove that the judgment describes identity in the world. Likewise, the equivalence relation generated from

non-breaking steps is a formal construction, not by itself a warrant for strict identity of the represented referent.

We select six reference kinds for analysis: obligation-bearers, rules, occurrences, scopes, records, and plain referents. The construction uses nine candidate carrier–basis profiles. Three pairs distinguish a locus from an object occupying it, a scope's coverage from its internal structure, and a rule's content from its structure. Forking, aggregation or decomposition, and refinement provide possible tests of those distinctions. Their proposed identity effects must be evaluated against the intended referent, the eligible presentations, and an independently justified identity criterion.

Within the selected construction, the declared classification functions are pairwise distinct, and the specified carrier–basis assignment is mathematically determinate. These are limited results. Distinct classification profiles need not generate distinct equivalence relations. The relationship depends on the generating rule, the admissible transformation dynamics, and the comparison domain (Appendix A).

Neither a difference between classification profiles nor a difference between generated relations independently establishes scientifically warranted strict identity. Nor does the selected catalog establish a minimum number of irreducible identity regimes. A carrier-indexed combination can preserve separate relations within one explicitly tagged relation, raising further questions about the meaning of "regime count" and the need for a non-amalgamation requirement.

The principal contribution is a conditional formal method for deriving and examining identity-preservation classifications under transformation.

First, identity claims become auditable: a verdict follows from declared identity invariants, admissible transformation steps, and explicit assumptions that can be independently examined.

Second, the construction supplies testable conditions for record-transforming systems. An admissible transformation step that preserves one declared in-

variant but changes another provides a concrete witness of different generated identity relations.

Third, the formalization identifies which assumptions carry each result. The scope and rule examples expose field-granularity and transformation-domain dependencies; the fork example exposes the need for explicit continuity warrants.

The proposed catalog remains a finite comparative construction. Its pairwise classification distinctions do not establish a necessary number of irreducible identity regimes.

These contributions establish conditional mathematical consequences, not independently warranted criteria for strict identity of real-world referents. Their scientific adequacy requires independent justification of the selected bases, transformation models, and warrants.

The remainder of the paper proceeds as follows. Section 2 states the Neutral Substrate foundation and the narrower contribution pursued here. Section 3 sets out the formal setting and transformation classifications. Section 4 specifies the selected reference inventory. Section 5 develops conditional comparisons among identity bases, and Section 6 collects the candidate profiles. Section 7 examines their distinctions and the limits of regime counting. Section 8 establishes determinacy of the selected profile assignment, while Section 9 treats three application settings as identity-criterion tests. Section 10 situates the work in prior research; Section 11 states its boundaries; and Section 12 concludes.

2 Background: Neutral Substrates

This paper builds on *Neutral Substrates* (Case 2026a). That work studies a shared representational base intended to support reference under persistent interpretive disagreement without adopting contested causal or normative interpretations as unqualified foundational commitments. The present paper

investigates proposed identity conditions and their behavior when representations are transformed. It does not assume that the abstract referential interfaces have been given adequate identity criteria for every intended kind of referent.

The three definitions below retain their SE-100 labels and wording. They establish the terminology imported for the present analysis; they do not independently justify a concrete referential regime.

2.1 Inherited Definitions

Definition 2.1 (Substrate)**.** A *substrate* $\mathcal{S}$ is a shared representational base providing stable reference for entities, occurrences, and institutional artifacts across a class of admissible interpretive frameworks $\mathbb{F}$. Stable reference consists of individuation, co-reference, and persistence.

A substrate is the foundational layer, not the complete record system. It can support later causal, normative, evidentiary, or institutional interpretations without adopting those interpretations as its own object-level commitments. The distinction between a referent and interpretations attributed to it is central; preserving a shared label or descriptive record does not itself establish that the intended referent has been fixed.

Definition 2.2 (Referential Regime)**.** A *referential regime* is the triple of

1. individuation conditions,
2. co-reference conditions, and
3. persistence conditions

by which a substrate fixes and tracks entities, occurrences, and institutional artifacts.

Individuation concerns what counts as one whole and sameness across presentations; co-reference concerns whether references denote the same referent; persistence concerns strict identity across settings and must be distinguished from succession, derivation, or counterpart relations. The abstract triple supplies interfaces for those questions, not their independently warranted answers. A relation defined by closure of proposed preserving steps does not, without additional grounding, establish identity in an intended domain.

Definition 2.3 (Referential Commitments)**.** Let $\mathcal{S}_{\text{ref}}$ denote the *referential commitments* of $\mathcal{S}$. These are the substrate-layer commitments fixed by its referential regime, including identifiers; the typing of entities, occurrences, and institutional artifacts; timestamps; provenance; and referential relations among them.

Identifiers, types, timestamps, provenance, and relations may supply materials for referential fixing. Their mere presence does not justify which object-language propositions are fixed by a proposed referential regime. That connection requires independently grounded interpretation.

Neutral Substrates distinguishes a permitted attribution from the proposition attributed: accepting the attribution does not require accepting its content. It also defines interpretive non-commitment, extension stability, semantic neutrality, and neutrality by design.

Its central result is conditional sufficiency. Under Referential Common Ground, restricting foundational commitments to referential commitments and permitted attribution propositions suffices for semantic neutrality relative to every admissible framework. Permitted attributions are determined by the referential commitments.

The result does not establish that Referential Common Ground holds in a given realization, that the adopted referential criteria are adequate, that neutrality has been guaranteed by a permitted design-time basis, or that the foundational restriction is necessary for every neutral architecture. In particular, the formal restriction alone does not show that no object-level

causal or normative proposition is contained in the purported referential commitments. Those boundaries also apply to the present paper.

2.2 Contribution

This paper selects a finite reference inventory, candidate identity bases, and transformation families through which to investigate identity-sensitive distinctions. It studies classifications of re-expression, annotation, refinement, aggregation or decomposition, re-contextualization, reassignment, substitution, branching, provenance extension, and state evolution. The resulting classification profiles support limited formal comparison; their empirical or semantic adequacy remains an independent question.

Definition 2.4 (Transformation-Invariant Identity)**.** Relative to a specified referent kind, proposed identity basis, and admissible transformation instances, a presentation is treated as *transformation-invariant* along steps classified as preserving the declared basis. A transformation classification expresses a proposed identity judgment relative to that basis; it does not, without independent justification, establish strict identity of the represented referent across those steps.

Section 3 formalizes candidate profiles and a relation generated from non-breaking steps; Section 5 compares selected basis-sensitive classifications. The counting question and the need for independent scientific warrant are examined in Sections 7 and 11.

3 Formal Setting

This paper investigates candidate conditions for stable reference under transformation in settings of persistent interpretive disagreement. Its formal results are relative to the selected carriers, identity bases, transformation families, and assumptions stated below. The construction does not inde-

pendently establish that those choices are scientifically adequate for every intended referent or that the selected profiles constitute a minimum number of irreducible identity regimes.

3.1 Persistent Disagreement

We consider environments in which interpretive frameworks may remain in disagreement and their extensions need not be coordinated. Frameworks may advance incompatible causal accounts, normative attributions, or interpretive classifications without being required to converge. The neutrality constraint of Section 2 concerns which propositions the shared substrate itself adopts; it does not adjudicate among those interpretations.

This setting assumes neither that all admissible frameworks agree nor that they are mutually consistent. The conditional neutrality result of *Neutral Substrates* also requires Referential Common Ground: the complete adopted referential commitment set must remain jointly compatible with each admissible framework. That premise is not established by the present inventory or transformation classifications.

Revising a substrate merely to make conflicting interpretations converge would not resolve their disagreement neutrally. Revising an inadequately justified referential commitment is a different matter: any revised construction requires renewed examination of its own compatibility and referential warrant. This paper does not define such a revision procedure.

3.2 Stable Reference in Strong Form

We examine the demanding case in which representations of a purported referent change through ordinary record transformations while the question of its identity remains meaningful. The referential regime in *Neutral Substrates* supplies abstract individuation, co-reference, and persistence interfaces; it does not independently determine which referents persist through a particular

operation.

Stable reference requires identifying the intended referent and supplying an adequate criterion for when presentations concern that same referent. A change in location, role, description, or state need not constitute a change of bearer identity; conversely, preservation of a record or label does not establish the persistence of what it represents. The applicable judgment depends on the proposed identity criterion and its independent justification.

Roles, flags, contextual predicates, and carrier/basis parameters are not inherently incompatible with neutral shared reference. They may carry explicit, fixed distinctions. What matters is whether their meanings and use are adequately specified and whether the resulting referential commitments satisfy the relevant common-ground conditions. A convention whose identity-relevant meaning is left to conflicting interpretations cannot, without further grounds, establish a shared identity judgment.

3.3 Transformation Basis

Identity questions become visible when representations are transformed. We select the following finite transformation basis for analysis:

$$\mathcal{F} = \{\mathrm{RE}, \mathrm{AN}, \mathrm{RF}, \mathrm{AD}, \mathrm{RC}, \mathrm{RA}, \mathrm{SU}, \mathrm{BF}, \mathrm{PV}, \mathrm{SE}\}.$$

The basis is a selected set of operation families, not a claim of completeness for record systems or of identity effects uniform across all referent kinds.

Definition 3.1 (Transformation basis)**.** The transformation basis $\mathcal{F}$ consists of these operation families:

(a) RE, re-expression: change of representation format while preserving represented content;
(b) AN, annotation: addition of non-identity-bearing information;
(c) RF, refinement: increase in descriptive, structural, or normative detail;
(d) AD, aggregation or decomposition: restructuring into coarser or finer

components under a specified whole-preservation condition;

(e) RC, re-contextualization: change in applicability context;
(f) RA, reassignment: change in association, bearer, or assignment relation;
(g) SU, substitution: replacement of the designated carrier;
(h) BF, branch or fork: divergence into multiple continuations;
(i) PV, provenance extension: addition of historical, derivational, or evidentiary metadata;
(j) SE, state evolution: change in state of an enduring referent over time.

These names designate typed operation families. Their meaning and identity effects must be evaluated relative to an intended referent, eligible presentations, and a stated identity basis. In particular, AD may preserve the covered cases of a scope while changing its internal organization; whether either property is identity-relevant is a separate question. Changes to covered cases are not represented by that particular whole-preserving instance of AD.

If a larger transformation basis preserves the previously assigned classifications on $\mathcal{F}$, then any existing classification disagreement remains a disagreement on the larger basis, as expressed in Proposition 7.2. This does not establish a minimum number of identity regimes.

3.4 Classification of Transformations

The construction assigns a classification to each operation family relative to a declared identity basis and its intended comparison domain.

Definition 3.2 (Transformation classification)**.** For a profile τ and transformation family $f \in \mathcal{F}$,

$$\mathrm{Class}_\tau(f) \in \{\mathsf{PRS}, \mathsf{BRK}, \mathsf{NEU}, \mathsf{N/A}\}.$$

The proposed classifications mean:

(a) PRS: the operation preserves the declared identity basis;
(b) BRK: the operation breaks the declared identity basis;

(c) NEU: the operation is applicable but does not affect that basis;
(d) N/A: the operation is inapplicable on the selected domain.

The four-valued classification above describes the stipulated profile catalog. The canonical Lean matrix uses three classification values (IGN, PRS, BRK), mapping both NEU and N/A to IGN. The companion Persistence classification restores applicability at the N/A cells of Table 1.

The newer invariant-based construction instead derives one of five verdicts: N/A, BRK, MIX, PRS, or NEU. It also distinguishes a modeled N/A verdict from a matrix cell for which no derivation has been supplied.

These representations are not interchangeable without an explicit semantic correspondence.

PRS and NEU both designate non-breaking behavior in this classification scheme, but they express different reasons. A family-level value is meaningful only when its proposed behavior is sufficiently uniform over the specified eligible operations; otherwise the intended scope must be refined. Neither the classification label nor its recorded presence in a profile independently warrants strict identity for particular referents.

Definition 3.3 (Non-breaking transformation)**.** A family f is *non-breaking* for profile τ when

$$\mathrm{Class}_\tau(f) \in \{\mathsf{PRS}, \mathsf{NEU}\}.$$

It is *breaking* when $\mathrm{Class}_\tau(f) = \mathsf{BRK}$.

3.5 Identity Regimes and Generated Relations

An identity regime is a proposed pairing of an identity basis with a transformation classification. It is not itself an ontological kind or proof that its basis is adequate for a particular referent.

Definition 3.4 (Identity regime)**.** An *identity regime* τ consists of a declared

identity basis and a classification function

$$\mathrm{Class}_\tau : \mathcal{F} \to \{\mathsf{PRS}, \mathsf{BRK}, \mathsf{NEU}, \mathsf{N/A}\}.$$

The basis proposes conditions of sameness for the intended referent; the classification records proposed transformation behavior relative to that basis.

Definition 3.5 (Induced relation from non-breaking families)**.** Fix a profile τ and a domain D_τ of eligible presentations. Let $E_\tau \subseteq D_\tau \times D_\tau$ consist of pairs joined by specified, admissible steps of transformation families f satisfying

$$\mathrm{Class}_\tau(f) \in \{\mathsf{PRS}, \mathsf{NEU}\}.$$

Define $\sim_\tau$ as the smallest equivalence relation on D_τ containing E_τ.

This is a classification-based family-level construction. It selects admissible steps according to the stipulated family classifications; it does not independently establish that each step preserves the declared identity basis.

The companion Persistence construction defines a different family-level relation, generated only by steps of families classified PRS. Under common domains and transformation dynamics, the PRS-only generated relation is contained in the relation generated by PRS and NEU families.

The newer invariant-based construction instead evaluates preservation at the level of individual admissible steps. Its generating edges are exactly the steps that preserve the declared invariant, including preserving steps of families whose derived verdict is MIX. The three constructions must not be conflated.

Each generated relation is an equivalence relation by construction. None independently establishes strict identity of the intended real-world referents. Such an interpretation requires an independently justified identity criterion, evidence supporting the admissible steps, and justification that equivalence closure preserves the criterion.

In particular, branching or a chain of classified non-breaking steps cannot, by itself, discharge these scientific obligations.

3.6 Basis Plurality

A carrier kind may admit more than one proposed identity basis. For example, one can compare locus-fixed with object-fixed criteria, coverage with structural organization, or rule content with rule structure. These comparisons are meaningful even before scientific adequacy and non-amalgamation are established.

Definition 3.6 (Basis plurality)**.** A carrier kind exhibits *basis plurality* within a construction when at least two distinct identity bases are specified for it. Whether both bases are independently justified, required for the intended applications, or irreducible is a separate question.

The selected construction examines three proposed pairs: plain referents tracked by locus or object, scopes by extension or structure, and rules by content or structure. Distinct descriptions or classification values do not alone prove distinct warranted identity relations.

3.7 Hidden Regimes

A representation can fail to disclose an identity-relevant distinction on which its identity judgments nevertheless depend. This issue is separate from whether those judgments are expressed in one relation or several relations.

Definition 3.7 (Hidden regime)**.** A *hidden regime* is an identity-relevant criterion or distinction used to determine sameness in a realization but not adequately declared or fixed in the shared referential account, so that its application depends on an unstated or contested interpretation.

A declared role, flag, carrier tag, or basis parameter is not hidden merely because it is a discriminator. Its adequacy depends on whether its semantics are fixed and its use is justified for the intended referents. Conversely, an unacknowledged convention cannot be treated as established common ground merely because a system implements it. Whether a proposed combined

representation is faithful must therefore be examined rather than decided by terminology alone.

3.8 Scope of the Analysis

The selected carrier inventory, identity bases, and transformation classifications supply a finite candidate construction. Sections 5–7 analyze its proposed transformation distinctions and the limits of the resulting count.

The analysis does not establish a universal minimum number of identity regimes, a complete inventory of identity criteria, or a general procedure for determining strict identity under transformation. Such conclusions would require justified referent-specific criteria, sound classifications, and a satisfactory account of faithful combinations and non-amalgamation.

The next section presents the selected reference inventory. Its carrier kinds identify subjects of inquiry; they do not themselves supply independent identity warrants.

4 Reference Inventory

Before analyzing identity under transformation, we select a finite inventory of referents encountered in the accountability settings considered in this paper. These settings call for descriptions of parties, rules, occurrences, scopes, records, and things acted upon or observed. The inventory organizes questions of reference without adopting a contested causal or normative interpretation as a substrate commitment.

The inventory is not a theorem, an upper ontology, or a complete domain ontology. It does not establish that all neutral substrates require these six kinds, that the kinds are mutually exclusive, or that their names supply adequate criteria of individuation or persistence. Their suitability for particular referents requires independent examination.

Reference kinds organize the intended referents; identity bases propose conditions for their sameness under transformation. Section 7 examines this counting question separately.

4.1 Reference Kinds and Identity Carriers

Definition 4.1 (Reference kind)**.** A *reference kind* is a class of intended referents selected to organize questions of reference in the accountability settings under analysis. Membership in a reference kind does not itself determine a referent's unity, sameness, or persistence criteria.

Definition 4.2 (Identity carrier)**.** An *identity carrier* is an intended referent considered as the subject of a proposed sameness, difference, or persistence judgment under transformation. The designation does not itself warrant that judgment or determine the applicable identity basis.

A reference kind can organize several proposed identity bases. Different kinds may also support compatible criteria or share referents considered in different capacities. In either case, the adequacy of an identity judgment depends on an identified bearer and an independently justified criterion, not solely on a kind name or transformation label.

The selected inventory is:

Reference kind	Referent considered in this analysis
Obligation-bearer	a party considered as a possible bearer of responsibility, duty, liability, or authority
Rule	a rule or normative instrument considered separately from an act of enacting, interpreting, or applying it
Occurrence	a happening such as a filing, inspection, transfer, or decision, whose unity and identity require specified criteria
Scope	a domain of applicability described by places, parties, periods, procedures, classifications, or conditions
Record	a document, entry, or other informational record that can be referred to independently of what it describes or asserts
Plain referent	a site, object, facility, instrument, sample, or asset considered apart from any assigned obligation-bearing or rule-governing role

These categories draw on distinctions familiar in foundational ontology, legal ontology, provenance, and institutional recordkeeping. No novelty or independent necessity is claimed for this inventory. Its role is to provide candidate carriers for the subsequent analysis.

4.2 Obligation-Bearers

The selected settings include claims about parties that may bear responsibility, duty, liability, authority, or accountability. The substrate can preserve reference to a party and record that a source attributes an obligation to it without adopting that attribution as an unqualified normative conclusion.

In multi-jurisdiction regulatory reporting, frameworks may identify different

entities or capacities as relevant: a parent organization, subsidiary, facility, office, or delegated agent. The organizational entity, an institutional role, and the occupant of that role need not have identical persistence criteria. Whether two presentations concern the same bearer requires an account appropriate to the intended referent.

Distinguishing a bearer from occurrences in which obligations are assigned, contested, fulfilled, or discharged supports comparisons across time. One bearer may participate in several occurrences, and one occurrence may involve several bearers. These observations motivate separate reference questions; they do not prove that the corresponding identity relations cannot be represented in a common formal structure.

4.3 Rules

The selected settings include reference to rules and normative instruments that may be enacted, amended, delegated, interpreted, applied, challenged, or repealed. Recording such a referent does not require the substrate to declare that the rule is valid, binding, justified, or correctly interpreted.

Legislative history analysis distinguishes questions about a statute from questions about the enactment, amendment, agency action, or judicial decision associated with it. A statute may remain citable across those occurrences, but citation continuity alone does not settle whether successive texts, versions, or normative contents constitute the same rule.

The proposed content-fixed and structure-fixed bases examine different responses to refinement. Whether either basis supplies a warranted identity criterion depends on which rule referent is intended and what the transformation actually preserves.

4.4 Occurrences

The selected settings include occurrences such as filings, inspections, votes, transfers, transactions, observations, decisions, enactments, amendments, and enforcement actions. Their temporal realization and provenance can help identify them but do not necessarily determine where one occurrence begins or ends, which phases belong to one whole, or when two reports concern the same happening.

For example, jurisdictions may disagree about whether a filing constitutes compliance while referring to an occurrence they have independently identified as the same filing. Similarly, institutions may dispute the significance of an enactment or a measurement without thereby disputing its occurrence identity. In other cases, that identity itself may be contested and cannot simply be assumed as common ground.

An occurrence can be distinguished for analysis from its participants, the records describing it, and the rules invoked in its evaluation. Such analytic distinctions do not independently establish disjoint ontological domains or irreducible identity regimes.

4.5 Scopes

The selected settings include applicability domains described by places, parties, periods, procedures, classifications, or conditions. Scopes may be territorial, institutional, temporal, procedural, classificatory, jurisdictional, or conditional. Their boundaries and relevance may themselves be disputed.

In multi-jurisdiction reporting, a rule may be associated with different entities, facilities, transactions, or reporting periods. Tracking a scope separately in an analysis can permit comparison of coverage while holding the rule description fixed, or comparison of rule versions while holding coverage fixed. This is a useful modeling choice, not proof that every substrate must make scope an independent identity carrier.

The proposed extension-fixed and structure-fixed bases distinguish coverage from internal organization. A decomposition may preserve covered cases while changing organization; whether it preserves identity depends on the justified scope criterion, not solely on the operation's name.

4.6 Records

The selected settings include descriptive documents, dataset entries, observation records, provenance items, and other informational records. A substrate may record that a source asserted a claim without thereby adopting its causal or normative content.

In cross-institutional provenance, a record may be produced, received, annotated, copied, transferred, or reanalyzed. The institutions may disagree about what it shows. Whether successive presentations are of the same record nevertheless depends on the identity basis: content, a particular token, a version, or a custody-linked object may give different answers.

Distinguishing a record from what it describes permits separate questions about an assertion and its subject. It does not imply that annotation, copying, or transfer always preserves record identity, or that record identity is an irreducible regime.

4.7 Plain Referents

The selected settings also include things acted upon, regulated, located, transferred, observed, measured, or described: facilities, parcels, instruments, devices, environmental sites, physical objects, infrastructure components, samples, and assets.

The designation *plain referent* indicates how a referent is being considered in the present analysis, not a claim that the referent could never bear an obligation, occupy a role, or appear in another reference kind. A facility may be inspected; a parcel may fall within an applicability scope; an instrument

may produce a measurement record; and a sample may pass between institutions. Whether the referent persists through these operations remains a separate question.

The proposed locus-fixed and object-fixed bases distinguish questions about a place from questions about an object associated with that place. A branch or fork may provide a useful test of their different transformation responses, but does not by itself establish that they are two identity criteria for one and the same kind of bearer. The intended referents and eligible operations must be specified.

4.8 Summary

The selected reference kinds organize a finite set of candidate identity questions.

The distinction between a selected carrier and its proposed identity basis remains essential. The inventory does not determine a warranted identity relation, prove the necessity of each candidate basis, or establish the non-amalgamation of the resulting relations.

The next section compares proposed identity bases and transformation behavior for these selected carriers under the formal setting of Section 3.

5 Transformation-Invariant Identity

The selected reference inventory of Section 4 identifies candidate carriers for comparing identity judgments under transformation. This section examines how declared identity bases and classification profiles can differ over the transformation basis of Section 3. A classification records proposed behavior relative to a specified basis; it does not independently establish that two presentations have the same real-world referent. The scientific adequacy of the basis, the operation, and their interpretation remains a separate

obligation.

The inquiry draws on the distinction between reference kinds and identity conditions familiar in formal ontology (Guarino 1999; Guarino and Welty 2002). A kind name alone does not settle whether a transformation preserves its intended bearer. A stated identity basis helps formulate a test, but a warranted persistence judgment also requires evidence and an appropriate criterion for that referent.

5.1 Identity Bases

Definition 5.1 (Identity basis)**.** An *identity basis* is a stated criterion proposed for deciding when eligible presentations concern the same intended referent. It specifies the conditions whose preservation would warrant sameness under the interpretation in question. Its declaration alone does not establish that the criterion is adequate for the intended referent.

The selected construction considers three candidate bases associated with obligation-bearers, occurrences, and records, and three pairs of bases: content versus structure for rules, extension versus structure for scopes, and locus versus object for plain referents. These are proposed criteria for examining transformation behavior, not an exhaustive inventory of scientifically adequate identity conditions.

A transformation classification is meaningful only relative to the identity basis, the eligible comparison domain, and the actual effects of the operation. Different candidate bases may classify a selected operation differently; that does not by itself decide which basis is appropriate or whether the resulting judgments warrant strict identity.

5.2 Profiles and Collapse

Section 3 defines identity regimes as proposed basis–classification profiles and defines an equivalence relation $\sim_\tau$ generated from specified non-breaking steps. The generated relation and an independently warranted strict-identity relation must not be conflated.

Definition 5.2 (Collapse)**.** Fix a common comparison domain D of eligible presentations for two profiles τ and τ'. The profiles *collapse at the level of generated relations on* D when the restrictions of their generated relations agree:

$$\sim_\tau\big|_D = \sim_{\tau'}\big|_D.$$

Those generated relations are *distinct on* D if one relates an eligible pair that the other does not.

Distinct classification functions do not necessarily induce distinct generated relations. For example, a family assigned PRS under one profile and NEU under another is included under both profiles in the classification-based PRS-or-NEU construction. On common domains and transformation dynamics, its admissible steps therefore contribute generating edges to both relations.

Those stipulated assignments do not independently establish that every actual step preserves both declared invariants.

In the older family-generated construction, which uses only families classified PRS, distinct sets of preserving families need not yield distinct relations on arbitrary transformation dynamics, because the distinguishing families may have no relevant steps. In the Persistence construction, two pairs of profiles have identical preserving sets and so generate identical relations on every shared dynamics (Appendix A).

A stronger result holds in the invariant-based step construction: when an admissible step inside a common comparison domain preserves one basis invariant but changes another, the two step-generated relations differ. The endpoints of the separating step cannot be connected by any path of steps

that each preserve the second basis, because equality of invariant values is preserved along such a path (Proposition 6.5). This is a mathematical fact about the specified invariants and admissible steps, not independent scientific warrant for strict identity of the intended referents.

Definition 5.3 (Underdetermined kind)**.** A reference kind is *underdetermined at a transformation family f* within the selected construction when two proposed bases for that kind assign, on a shared eligible comparison domain, a non-breaking classification and a breaking classification, respectively, to f.

This is a difference between candidate classification profiles. If both bases are adequately justified for a particular use, a system must retain the distinction between the judgments they authorize. It does not follow that they require separately implemented or non-amalgamable relations: an explicit basis parameter may preserve both. Nor does a classification difference alone establish a separating pair in the generated equivalence relations or in the world.

5.3 Carrier Kinds Not Split in the Core Construction

The selected catalog specifies one candidate basis for each of three kinds: obligation-bearer, occurrence, and record. This selection does not establish that no other adequate bases exist.

For an *obligation-bearer*, a possible criterion tracks the continuing party rather than each obligation attributed to that party. A change in location, assigned role, or obligation may preserve that party, provided an appropriate account of the party's continuity warrants the judgment. Replacement of the party may instead break its identity. Organizational continuity and continuity of role occupancy must not be conflated.

For an *occurrence*, temporal realization and provenance may supply useful evidence, but they do not by themselves establish one bounded whole. Redescription of an independently individuated occurrence may preserve it;

treating a proper phase as the whole may not. Decomposition, branching, and other transformations cannot be classified uniformly without specifying the occurrence and eligible presentations.

For a *record*, content identity, token identity, version identity, and continuity of custody can produce different judgments. Copying a report may preserve its content while creating another token; annotation may preserve an informational core while creating a new version. Neither preservation nor breakage follows from the label "record" alone.

The selected profile names are

$$\text{OBL}, \qquad \text{OCC}, \qquad \text{REC}.$$

Their intended identity criteria and transformation classifications remain subject to the qualifications above.

5.4 Witnessed Split Kinds

The other three carrier kinds in the selected catalog have pairs of candidate bases. Their comparisons can be expressed by a common classification pattern. In particular, a difference between proposed values of $\mathrm{Class}_{\tau}(f)$ is an algebraic fact about the profiles; a difference between warranted identity relations requires more.

Lemma 5.4 (Split pattern for classification profiles)**.** *Let K be a reference kind and let τ_1, τ_2 be two candidate profiles for K over the same transformation basis and a shared eligible comparison domain. If some $f \in \mathcal{F}$ satisfies*

$$\mathrm{Class}_{\tau_1}(f) \in \{\mathsf{PRS}, \mathsf{NEU}\} \quad \textit{and} \quad \mathrm{Class}_{\tau_2}(f) = \mathsf{BRK},$$

then the classification functions of τ_1 and τ_2 are distinct. This conclusion alone does not establish distinct generated equivalence relations, scientifically warranted strict-identity relations, or a minimum number of irreducible

regimes.

Proof. The first classification is different from BRK, so the classification functions disagree at f and cannot be equal. Their generated relations depend on the eligible step pairs and equivalence closure; those relations, and their interpretations, are not determined by this disagreement alone. □

A stronger, referent-level separation would need a common eligible pair of presentations together with independently justified identity criteria that agree and disagree on that pair. The following examples show how candidate separating situations can be constructed. They do not prove that the corresponding criteria are necessary for every realization of the reference kind.

5.5 Scope Under Decomposition

For a scope presentation, distinguish the cases covered from their internal organization. Consider a finite set $C = \{a, b\}$ and two partitions

$$P = \{\{a, b\}\}, \qquad P' = \{\{a\}, \{b\}\}.$$

The presentations (C, P) and (C, P') have the same extension but different partition structures. A stipulated extension-fixed basis identifies them; a stipulated structure-fixed basis distinguishes them. An eligible decomposition that replaces P by P' while preserving C therefore provides a candidate separating example.

Proposition 5.5 (Scope underdetermination in the selected model)**.** *If an eligible* AD *operation preserves the coverage C while changing the identity-relevant partition P, then the extension-fixed and structure-fixed candidate profiles can assign different classifications to that operation.*

Proof. Under the stipulated extension-fixed criterion, equality of C supplies preservation; under the stipulated structure-fixed criterion, inequality of P

supplies breakage. On a domain where these are the specified effects of AD, its two classifications differ, as in Lemma 5.4. □

We call the two proposed profiles

SCOPE-E, SCOPE-S.

This example demonstrates a distinction between coverage and structure within the model. It does not show that partition equality is the correct identity criterion for every scope, or that a scope cannot be faithfully represented with an explicit basis parameter.

5.6 Rule Under Refinement

For a rule presentation, distinguish the specified normative content from its textual or organizational structure. Consider two presentations (N, S) and (N, S') with identical stipulated content N but different structures $S \neq S'$. A content-fixed comparison identifies the presentations, whereas a structure-fixed comparison separates them. Whether a particular refinement actually preserves normative content must be assessed rather than inferred from the word "refinement."

Proposition 5.6 (Rule underdetermination in the selected model)**.** *If an eligible* RF *operation preserves the specified content N while changing the identity-relevant structure S, then the content-fixed and structure-fixed candidate profiles can assign different classifications to that operation.*

Proof. The stipulated content-fixed criterion is preserved because N is unchanged; the stipulated structure-fixed criterion is broken because $S \neq S'$. The corresponding classifications differ, as in Lemma 5.4. □

We call the two proposed profiles

RULE-C, RULE-S.

This comparison does not decide whether a particular statute or rule persists across amendment, recodification, or re-expression. That requires a justified account of the relevant rule referent.

5.7 Plain Referent Under Forking

For plain referents, distinguish a locus from an object associated with that locus. A toy configuration (L, O) records both. Two configurations (L, O_1) and (L, O_2) may have the same locus L but different objects $O_1 \neq O_2$. A locus-fixed comparison then identifies them, while an object-fixed comparison distinguishes them. The inequality of the objects is an assumption of the toy model, not a conclusion from their labels alone.

Proposition 5.7 (Plain-referent underdetermination in the selected model)**.** *If an eligible* BF *operation yields compared presentations with the same locus but independently distinct objects, then locus-fixed and object-fixed candidate profiles can assign different classifications to that comparison.*

Proof. The locus-fixed criterion is preserved by the shared locus; the object-fixed criterion is broken by the stipulated object distinction. The candidate classifications consequently differ, as in Lemma 5.4. □

We call the two proposed profiles

LOC, OBJ.

A fork of record representations does not by itself create distinct objects or imply loss of object identity. Nor does relocation of an object necessarily end its identity. The separating case depends on which referent is being tracked and whether the assumed operation and object distinction are independently warranted.

5.8 Result of the Algebra

The selected inventory contributes three single-basis candidate profiles and three pairs of candidate profiles:

OBL, OCC, REC,
LOC, OBJ, SCOPE-E, SCOPE-S, RULE-C, RULE-S.

The conditional examples show how different identity bases can classify specified transformations differently. They do not establish that every transformation of the named family has the same effect, that the generated relations warrant strict identity, or that all selected profiles are scientifically necessary or non-amalgamable.

Section 6 collects the proposed classifications; Section 7 distinguishes the catalog count.

6 Candidate Carrier–Basis Profiles

The selected construction of Sections 4 and 5 contains nine named carrier–basis profiles. Three carrier kinds are represented by one candidate basis each, and three by pairs of candidate bases. This is a catalog selected for analysis, not a claim that all selected profiles are scientifically necessary or irreducible.

The following labels name the selected profiles:

OBL	obligation-bearer identity.
OCC	occurrence identity.
REC	record identity.
LOC	locus-fixed plain-referent identity.

OBJ	object-fixed plain-referent identity.
SCOPE-E	extension-fixed scope identity.
SCOPE-S	structure-fixed scope identity.
RULE-C	content-fixed rule identity.
RULE-S	structure-fixed rule identity.

Table 1 records proposed classifications over the finite transformation basis $\mathcal{F}$. Its entries are stipulated family-level profile assignments for the selected comparison settings. They must not be read as established identity effects of every operation in the named family or as independently warranted judgments about real-world referents. Where eligible operations within a family have different effects, that family requires a more specific classification domain.

	OBL	OCC	REC	LOC	OBJ	SCOPE-E	SCOPE-S	RULE-C	RULE-S
RE	NEU	NEU	NEU	NEU	NEU	NEU	NEU	NEU	NEU
AN	NEU	NEU	PRS	NEU	NEU	NEU	NEU	NEU	NEU
RF	PRS	PRS	PRS	PRS	PRS	PRS	BRK	PRS	BRK
AD	PRS	BRK	PRS	PRS	PRS	PRS	BRK	PRS	BRK
RC	NEU	NEU	NEU	NEU	NEU	PRS	BRK	NEU	NEU
RA	NEU	N/A	NEU	N/A	N/A	NEU	BRK	NEU	NEU
SU	BRK	BRK	BRK	BRK	BRK	BRK	BRK	BRK	BRK
BF	NEU	N/A	PRS	PRS	BRK	PRS	BRK	BRK	BRK
PV	NEU	NEU	PRS	NEU	NEU	NEU	NEU	NEU	NEU
SE	NEU	PRS	NEU	PRS	PRS	N/A	N/A	N/A	N/A

Table 1: Proposed classification profiles over $\mathcal{F}$ for the selected carrier–basis construction. PRS: preservation of the declared basis; BRK: breakage of that basis; NEU: applicable but identity-neutral under that basis; N/A: inapplicable on the selected domain. Entries are candidate assignments, not universal identity laws.

A PRS entry proposes preservation of the declared identity basis; a BRK entry proposes its breakage. An NEU entry indicates an applicable operation that does not affect the basis, and an N/A entry indicates inapplicability on the selected domain. Neither PRS nor NEU independently warrants strict identity between actual referents. In particular, a BRK entry in a proposed profile does not by itself establish that two specified objects are distinct.

For the selected same-carrier comparisons, BF distinguishes the proposed LOC and OBJ classifications; AD distinguishes SCOPE-E and SCOPE-S; and RF distinguishes RULE-C and RULE-S. The conditional examples in Section 5 show how stipulated locus, object, extension, structure, and content criteria can yield these classification differences. The other entries record additional proposed behavior whose adequacy also requires examination; they are not needed to establish the three conditional split comparisons.

The uniform RE and SU rows are features of this selected table: RE is assigned NEU throughout, while SU is assigned BRK throughout. Those assignments are not general theorems about re-expression or substitution under every scientifically adequate identity criterion.

A disagreement between two classification functions is distinct from a disagreement between the equivalence relations generated by non-breaking steps. To state the latter, the two relations must be compared on a common eligible domain.

Definition 6.1 (Separation witness)**.** Let τ and τ' be profiles with a common domain D of eligible presentations. A *separation witness for their generated relations on* D is a pair $(r, s) \in D \times D$ for which

$$r \sim_\tau s \quad \text{and} \quad r \not\sim_{\tau'} s,$$

or conversely, with each relation restricted to D.

Such a witness demonstrates that the generated relations differ on D; it does not independently establish the correctness of either relation as strict real-world identity. Nor does a difference in the classification of a transformation family necessarily provide such a witness, since equivalence closure may connect the relevant pair by another non-breaking path.

Proposition 6.2 (Distinct core profiles)**.** *The nine classification functions specified by the columns of Table 1 are pairwise distinct.*

Proof. It is enough to inspect the entries at AN, RF, RA, and BF. The

four-entry classification signatures in these rows are pairwise different across all columns of Table 1. Hence no two of the classification functions agree on all of $\mathcal{F}$. □

This proposition establishes distinctness of the *declared classification functions* in the selected catalog. It does not establish pairwise distinctness of their generated relations on a common domain, that all candidate identity bases are independently warranted, or that any faithful realization requires a necessary number of regimes. Those questions are examined as limits of the counting argument in Section 7.

6.1 Initial Model-Derived Classification Results

Table 1 records the proposed classification matrix. Its entries are not inputs to the following derivations.

Instead, the companion Lean models (Case 2026b) specify presentation fields, identity invariants, admissible transformation steps, and their effects.

An invariant basis consists of a function on presentations and a declared support of presentation fields. Agreement on the supported fields implies equality of the invariant. The Lean development additionally proves minimality of the supports used in the certified cells: every supported field can independently affect its invariant.

An effect model declares admissible transformation steps and the fields each transformation family may write. An effective instance is an admissible step that changes at least one presentation field.

Definition 6.3 (Model-derived transformation verdict)**.** Fix an invariant basis b, an effect model M, and a transformation family f.

The derived verdict is:

(a) N/A: no effective instance exists;
(b) BRK: effective instances exist and every one changes the invariant;

(c) MIX: some effective instance changes the invariant and another preserves it;
(d) PRS: effective instances exist, all preserve the invariant, and at least one changes a supported field;
(e) NEU: effective instances exist, all preserve the invariant, and none changes a supported field.

MIX establishes that the declared transformation family contains both invariant-preserving and invariant-breaking effective instances.

The family therefore cannot receive a uniform preserving or breaking verdict under those modeled conditions. A uniform verdict would require restricting or refining the family.

Alternatively, individual preservation judgments can be retained at the step level without assigning one uniform verdict to the family.

Applicability is defined for the model's presentation space, not for an arbitrarily selected smaller comparison domain.

The certified cells in the derived matrix of the companion Lean development are:

- SCOPE-E under strict AD: NEU;
- SCOPE-S under strict AD: BRK;
- RULE-C under strict RF: NEU;
- RULE-S under strict RF: BRK;
- LOC under BF with Colocated and FreshObject warrants: NEU;
- OBJ under BF with those same warrants: BRK.

Each certified cell contains an explicit effect model, an invariant basis with a proven minimal support, and a proof of its derived verdict. The remaining cells in the stipulated matrix have no derived model yet. For every certified cell, the derived and stipulated classifications agree on whether the transformation is breaking.

Pair and family	Model or explicit warrant	First	Second
SCOPE-E/SCOPE-S, AD	Coverage-preserving; one organization field	PRS	BRK
SCOPE-E/SCOPE-S, AD	Coverage-preserving; field per case-pair	NEU	BRK
SCOPE-E/SCOPE-S, AD	Broader variant allowing coverage changes; field per pair	MIX	BRK
RULE-C/RULE-S, RF	Norm-preserving strict refinement; coarse organization field	PRS	BRK
RULE-C/RULE-S, RF	Norm-preserving strict refinement; fields per case-pair and case norm	NEU	BRK
RULE-C/RULE-S, RF	Broader variant permitting changes to norms	MIX	BRK
LOC/OBJ, BF	Branch changes; no further warrant	MIX	MIX
LOC/OBJ, BF	Colocated and FreshObject warrants	NEU	BRK
LOC/OBJ, BF	Colocated and SameObject warrants	NEU	NEU

Table 2: Selected verdicts under explicitly specified Lean effect models and warrants. The two rightmost columns follow the order of the bases named in the first column. The field-level strict AD, strict RF, and warranted BF rows account for the certified cells in the derived matrix. The remaining rows are supplementary model comparisons. All verdicts are conditional mathematical results, not universal identity laws.

The three derived NEU cells are classified PRS in Table 1. Both verdicts are non-breaking, but they distinguish whether a transformation engages a supported field.

The scope and rule comparisons expose dependencies on field granularity. The fork classification additionally depends on explicit continuity warrants. These conditional results do not establish a universal correspondence between stipulated and derived classifications.

The additional rows in Table 2 are sensitivity analyses or alternative warranted models, not additional certified matrix cells.

Two findings are especially consequential. First, refining the scope presentation from one organization field to one field per case-pair changes the strict AD extension verdict from PRS to NEU. The fine-grained extension

basis reads diagonal entries (coverage), while restructuring changes only off-diagonal entries. The support of each fine-grained scope basis is proved minimal in that model. Both verdicts are non-breaking, but the classification is not invariant under the choice of field representation. The model therefore exposes a granularity dependence that must be controlled before an unqualified classification can be asserted.

The rule model exhibits a corresponding dependence. Under strict RF, the content basis is PRS when the organization is represented coarsely, but NEU when fields separately represent case-pair organization and case-specific norms. The structure basis is BRK in both representations. The field-level content support reads diagonal organization entries and the case-specific norms; strict RF preserves both. The content and structure supports are proved minimal in this model.

Second, branching alone specifies a change of branch, not a change of locus or object. With no additional warrant, both corresponding verdicts are MIX. Under the Colocated and FreshObject warrants, the locus basis is neutral and the object basis breaking; under the Colocated and SameObject warrants, both are neutral and their step-generated relations collapse on every comparison domain. These outcomes are consequences of distinct, explicitly stated hypotheses. None establishes that one warrant is true of every fork in an actual system.

The broader scope variant in the table permits changes to coverage and therefore is not an instance of the coverage-preserving AD family as originally defined in Section 3. It is included as a separate sensitivity analysis. The broader rule variant likewise changes the specification of eligible refinements. A comparison with Table 1 is warranted only after matching operation semantics, identity basis, field granularity, and admissibility conditions.

The invariant-based step construction uses a presentation space C, an effect model M, an invariant basis b, and a comparison domain $D \subseteq C$.

Define the preserving-step relation by

$$R_{M,b,D}(x,y) \iff x,y \in D \ \wedge \ \exists f \, (M.\mathrm{step}(f,x,y) \wedge b.\mathrm{inv}(x) = b.\mathrm{inv}(y)).$$

The step-generated identity relation is

$$\mathrm{StepId}_{M,b,D} = \mathrm{EqCl}_C(R_{M,b,D}),$$

where EqCl_C is the smallest equivalence relation on C containing the specified generating steps.

Unlike the classification-based family-level constructions, this relation tests invariant preservation for each admissible step. Its generating edges may include preserving steps from families whose derived verdict is MIX.

Theorem 6.4 (Step-generated relation equality)**.** *Fix a common effect model, two invariant bases b and b', and a comparison domain D.*

The equivalence relations generated by their respective invariant-preserving admissible steps are equal if and only if the bases agree on invariant preservation for every admissible step with both endpoints in D.

Proof. If preservation judgments agree on every admissible step in D, the generating relations are equal. Their equivalence closures therefore agree.

Conversely, suppose the generated relations agree. An admissible step preserving b belongs to the relation generated by b and hence also to the relation generated by b'.

Every generating step for the latter relation preserves b'. Equality of invariant values is an equivalence relation, so its closure also preserves b'. Thus the original step preserves b'. The converse follows symmetrically. □

Proposition 6.5 (Separating transformation step)**.** *Let $x, y \in D$ be connected by an admissible transformation step.*

If $b(x) = b(y)$ but $b'(x) \neq b'(y)$, then the step-generated identity relations for b and b' are distinct.

Proof. The step generates a relation between x and y under b.

Under b', the generated relation cannot connect x and y, because every generating step preserves b', and equality of invariant values is preserved under equivalence closure. □

These derivations are mathematically conditional.

Independently justifying the invariants and fork warrants for actual referents, and showing that a generated equivalence relation expresses warranted strict identity, remain separate scientific obligations.

7 Regime Distinctions and the Limits of Counting

The preceding sections specify a finite inventory of carrier kinds, proposed identity bases, and transformation classifications. They exhibit distinctions within that construction. A separate question is whether every realization preserving those distinctions must implement a particular minimum number of irreducible identity regimes.

7.1 The Inventory and Its Identity Conditions

Section 4 selects six carrier kinds for the accountability setting considered here: obligation-bearers, rules, occurrences, scopes, records, and plain referents. To reason about stable reference to any of these carriers, a realization needs adequate criteria for the relevant referents and the identity judgments it makes about them. The selection of these kinds is an assumption of the chosen inventory, not proof that all neutral substrates require them.

Nor does the distinction between carrier kinds establish a minimum number of irreducible identity regimes. Distinct representation domains can have distinguishable identity relations without those relations requiring separate, non-amalgamable formal structures. A lower-bound claim must first say what counts as one regime and which ways of combining relations are admissible.

7.2 Witnessed Differences Within the Selected Profiles

The proposed core contains three pairs of identity bases associated with the same carrier kind. These pairs motivate tests of whether ordinary transformations make different identity judgments under the respective bases.

For plain referents, the proposed locus-fixed and object-fixed bases are compared using branch/fork operations (Proposition 5.7). A particular fork may preserve a locus while distinguishing the objects associated with that locus. The question for a realization is which bearer each basis tracks and whether the proposed preservation and breakage judgments are warranted.

For scopes, the proposed extension-fixed and structure-fixed bases are compared using aggregation and decomposition (Proposition 5.5). An operation may preserve coverage while changing component structure. Whether that difference is an identity difference depends on the independently justified criterion for the scope in question.

For rules, the proposed content-fixed and structure-fixed bases are compared using refinement (Proposition 5.6). A refinement may preserve specified content while changing structure. Again, the relevant sameness judgment depends on what the rule is and on which criterion of its identity is justified.

If two justified identity relations disagree on the same eligible pair of presentations, they are distinct relations on that comparison domain. A difference between declared transformation classifications alone, however, does not establish such a disagreement about real referents. It requires a sound in-

terpretation of the operation, its applicable presentations, and the asserted identity basis. The three pairs are therefore useful candidate distinctions, not independent demonstrations of universally required regimes.

7.3 Counting Profiles Versus Counting Relations

The profiles catalogued in Section 6 arise from the selected carrier–basis assignments. Their number is a property of the chosen construction, not an exhaustive inventory of admissible identity criteria.

Proposition 6.2 concerns distinctions among the specified classification profiles.

The number of potentially distinct family-level generated relations depends on the generating rule. The PRS-only and PRS-or-NEU constructions induce different family-set equality patterns. Their corresponding upper bounds are established in Appendix A.

Additional equalities may arise when the dynamics do not distinguish the selected family sets. Neither count applies as a general bound on the newer invariant-based step relations, nor does either establish a minimum number of scientifically warranted identity regimes.

Even when profile distinctions are formally established, the further claim that they cannot be represented together without loss is a separate obligation.

7.4 Carrier-Indexed Combination and the Counting Question

An elementary construction illustrates why distinct carrier kinds do not, without more, establish an irreducible regime count. Let K index the selected carrier kinds, and let D_k be a domain of presentations for carrier kind k. Suppose E_k is an equivalence relation on D_k for each $k \in K$. Use the tagged disjoint union

$$D = \bigsqcup_{k \in K} D_k$$

and define a relation E on D by

$$(k, x)\ E\ (\ell, y) \quad \Longleftrightarrow \quad (k = \ell\ \wedge\ x\ E_k\ y).$$

Proposition 7.1 (Carrier-indexed combination)**.** *Then E is an equivalence relation on D. Restricting E to the k-th summand recovers E_k for each $k \in K$.*

Proof. Reflexivity and symmetry follow from the corresponding properties of each E_k. If $(k, x)\ E\ (\ell, y)$ and $(\ell, y)\ E\ (m, z)$, then $k = \ell = m$ and $x\ E_k\ y\ E_k\ z$. Transitivity of E_k therefore gives $(k, x)\ E\ (m, z)$. For presentations in the k-th summand, the definition of E is exactly the definition of E_k. □

This construction preserves the component equivalence relations without identifying presentations belonging to different carrier kinds. It therefore blocks an inference from disjoint carrier domains alone to an irreducible count of identity relations. It does not prove that the combined relation is one scientifically adequate identity regime, that it preserves all required transformation semantics, or that it satisfies the Neutral Substrate constraint. In particular, it does not resolve the differing identity bases proposed for the same carrier kind.

The unresolved question is whether a faithful realization can combine the selected identity criteria while retaining every warranted identity judgment and transformation distinction. A publicly declared carrier or basis parameter is not automatically a hidden regime merely because it is a parameter. Conversely, naming a parameter does not establish that its semantics are adequate or that it preserves the required distinctions. The admissibility of such a combination, and any scientifically justified non-amalgamation condition, require further analysis.

Proposition 7.1 constructs a tagged union preserving the component equivalence relations.

The outstanding question is whether a proposed combined realization also preserves all warranted identity judgments, transformation distinctions, and required referential commitments.

7.5 Separation Under an Extended Transformation Basis

A narrower monotonicity result applies to the declared classification profiles. Let $\mathcal{F} \subseteq \mathcal{F}'$ be transformation bases. Suppose the classifications of two profiles on $\mathcal{F}'$ agree with their existing classifications on every operation in $\mathcal{F}$.

Proposition 7.2 (Persistence of a classification distinction)**.** *If two profiles have different classifications for some $f \in \mathcal{F}$, their classification functions remain distinct under extensions to $\mathcal{F}'$ that preserve the original classifications.*

Proof. The extended classification functions still disagree at f, so they cannot be equal. □

This result concerns the preservation of a previously recorded classification difference. It does not establish a minimum number of irreducible identity regimes, nor does it establish that a classification difference warrants strict identity or non-identity for particular referents.

7.6 Boundary of the Result

The present analysis specifies a finite set of candidate carrier–basis profiles, identifies conditional transformation-level distinctions, and establishes formal consequences of the stated classification and relation constructions. The selected carrier inventory and identity bases require independent scientific justification for their intended referents.

A difference between classification profiles does not necessarily yield distinct generated relations or scientifically warranted identity judgments.

The carrier-indexed construction demonstrates that separate equivalence relations can be represented together. Whether such a combination preserves all warranted identity judgments, transformation distinctions, and required referential commitments remains an open scientific question. Any restriction on admissible combinations requires independent justification.

The broader limitations are discussed in Section 11.

8 Determinacy of the Selected Profile Assignment

Sections 4–6 specify a finite catalog of carrier–basis pairs and proposed transformation-classification profiles. This section asks whether the catalog assigns a named profile to each selected pair without ambiguity. This is an internal property of the specified assignment, not a claim that a particular identity basis is scientifically adequate, that the catalog is exhaustive, or that a minimum number of irreducible regimes are required.

8.1 Core Carrier–Basis Pairs

Definition 8.1 (Core carrier–basis pair)**.** A *core carrier–basis pair* is one of the nine selected carrier-kind and candidate identity-basis combinations displayed below. The arrows record the profile names assigned within this

construction:

$$
\begin{aligned}
\text{obligation-bearer with responsibility-bearing identity} &\mapsto \text{OBL},\\
\text{occurrence with realization-and-provenance identity} &\mapsto \text{OCC},\\
\text{record with descriptive-record identity} &\mapsto \text{REC},\\
\text{plain referent with locus-fixed identity} &\mapsto \text{LOC},\\
\text{plain referent with object-fixed identity} &\mapsto \text{OBJ},\\
\text{scope with extension-fixed identity} &\mapsto \text{SCOPE-E},\\
\text{scope with structure-fixed identity} &\mapsto \text{SCOPE-S},\\
\text{rule with content-fixed identity} &\mapsto \text{RULE-C},\\
\text{rule with structure-fixed identity} &\mapsto \text{RULE-S}.
\end{aligned}
$$

The first three carrier kinds have one candidate basis each in this catalog. The remaining three have two candidate bases each, motivated by the conditional comparisons in Section 5. The selection does not establish that these bases are necessary, sufficient, or the only adequate bases for those carrier kinds.

Definition 8.2 (Core regime assignment)**.** The *core regime assignment* A maps each selected core carrier–basis pair to the named classification profile indicated by the corresponding arrow in Definition 8.1. Its codomain is the selected profile catalog in Section 6.

The assignment is stipulated as part of the finite construction. It does not choose or justify a basis for an actual referent.

8.2 Existence

Lemma 8.3 (Assignment existence)**.** *Every selected core carrier–basis pair has an assigned profile under A.*

Proof. Definition 8.1 lists all selected pairs and gives an image for each. By Definition 8.2, A is defined on every pair in its stated domain. □

8.3 Uniqueness

Lemma 8.4 (Assignment uniqueness)**.** *Each selected core carrier–basis pair has exactly one assigned profile under A.*

Proof. Each pair occurs exactly once on the left-hand side of the displayed assignment in Definition 8.1, with one profile name on the right-hand side. The definition of A assigns no additional value to that pair. Hence its image under A is unique. □

8.4 Determinacy Theorem

Theorem 8.5 (Core regime assignment)**.** *For each carrier–basis pair in the selected catalog, there exists exactly one named classification profile assigned to it by A.*

Proof. Existence is Lemma 8.3; uniqueness is Lemma 8.4. □

This theorem records determinacy of a defined, finite assignment. It does not derive the assignment from the neutrality constraint, establish the scientific correctness of its classification entries, or show that the assigned profile yields a warranted strict-identity relation on real-world referents. Other candidate profiles or identity bases may be relevant outside this selected catalog.

8.5 Limits of Basis Declaration

A carrier-kind label alone does not select one profile when the catalog contains more than one candidate basis for that kind. For example, a scope may be examined under extension-fixed or structure-fixed criteria, a rule under content-fixed or structure-fixed criteria, and a plain referent under locus-fixed or object-fixed criteria. Supplying the corresponding basis label makes the input to A unambiguous *within this catalog.*

That is not the same as establishing which basis is appropriate for a particular referent. A declared basis needs independently justified individuation and persistence conditions, and each proposed transformation classification needs an adequate interpretation. Even the carrier kinds represented by only one profile in this catalog may admit other legitimate identity bases.

Consequently, a possible conformance check could verify whether a system discloses a selected basis and applies its declared profile consistently. It could not, on that evidence alone, certify the basis's scientific adequacy or the truth of its identity judgments. This is the boundary of the determinacy result proved here.

9 Applications as Identity-Criterion Tests

Sections 4–8 define a selected inventory, candidate carrier–basis profiles, and their stipulated transformation classifications. The following settings illustrate questions that those distinctions can help formulate. They do not establish that the proposed bases are scientifically adequate for every referent, that the corresponding generated relations warrant strict identity, or that a minimum number of irreducible regimes are necessary.

For each setting, the relevant questions are which referent is intended, which presentations are eligible for comparison, what changes under a transformation, and what evidence justifies sameness or difference. Where a common referential judgment cannot be warranted, the Neutral Substrate construction does not license adopting it as an unqualified foundational commitment merely because the records share a label.

9.1 Multi-Jurisdiction Regulatory Reporting

A regulatory record may refer to a corporation, a subsidiary, an office, or a party occupying an office. The selected OBL profile offers a candidate

basis for following an obligation-bearer across changes in reported duties and activities. Whether the same legal or institutional bearer persists requires an appropriate continuity criterion; sharing a name, role, or identifier is not sufficient. Different jurisdictions may also identify different bearers for a purported obligation. The shared record must distinguish an identified party from a framework's attribution of responsibility to that party.

A reporting statute is a candidate rule referent. Under RULE-C, the comparison asks whether specified normative content is preserved; under RULE-S, it asks whether a specified textual or internal structure is preserved. Recodification might preserve requirements while changing section organization, yielding different *proposed* classifications. That comparison does not decide whether the resulting enactment is the same statute as a legal object. Its legal continuity, the meaning of its requirements, and the intended unit of reference must be justified independently.

Filings, inspections, transactions, and enforcement actions provide candidate occurrences for OCC. A filing occurrence is distinguishable from the filed document and from subsequent copies or versions of that document, which can be examined under REC. Two jurisdictions may agree which filing occurred while disagreeing whether it constituted compliance. They may also disagree about whether two reported activities are the same occurrence. In the latter case, common occurrence identity cannot simply be assumed from matching timestamps or document identifiers.

A jurisdictional scope can be examined by the cases it covers (SCOPE-E) or by a specified administrative or procedural structure (SCOPE-S). Dividing a reporting domain into districts may preserve its coverage while changing its partition. This supplies a conditional transformation comparison, not a general rule that the jurisdiction itself persists under subdivision. The identity of a legal jurisdiction may depend on further institutional facts not captured by either profile.

A regulated facility, parcel, or instrument supplies a possible plain referent. The LOC comparison follows a site or location, whereas OBJ follows the

object occupying or moving through it. A facility relocated from one site to another need not cease to be the same facility; conversely, continued use of one site does not make successive occupants one object. The proposed basis must match the intended bearer.

These comparisons make all selected carrier kinds available for analysis and illustrate the three candidate split pairs. They do not show that every regulatory system needs every one of the candidate profiles. Nor does agreement about a carrier's designation establish the identity criterion required to compare it across jurisdictions.

9.2 Legislative History Tracking

A statute can be tracked by its recorded requirements, its enacted text, its section structure, or the legal continuity of the instrument across amendments. The selected RULE-C and RULE-S profiles isolate content- and structure-based comparisons. An amendment may retain specified requirements while reorganizing sections; a textual continuation may also be assigned a changed interpretation. Neither observation alone establishes the strict identity of the statute before and after the amendment. The referent and its identity conditions must first be fixed.

Enactments, votes, amendments, applications, and judicial decisions are candidate occurrences for OCC. An enactment is not the enacted rule, and an account of an enactment is not the enactment itself. Legislative texts, committee reports, docket entries, and interpretive memoranda supply record candidates for REC. Copies with identical content may be distinct documentary tokens, and later versions may differ without settling whether they remain the same informational object. The record basis must therefore be specified rather than presumed.

Legislatures, agencies, courts, officers, and regulated parties can be considered as obligation- or authority-bearing parties under OBL, provided the intended bearer is distinguished from an office, its occupant, or a particular

institutional role. The same office may have successive occupants; the same person may move to another office. Such changes do not, by themselves, settle the identity of either the person or the office.

Geographic, procedural, temporal, and institutional boundaries provide candidate scopes. A redistricting operation can preserve a chosen set of covered cases while changing a partition, offering a comparison between SCOPE-E and SCOPE-S. Objects affected by the legal regime, such as land parcels, facilities, or instruments, may be considered under LOC or OBJ when those bases are appropriate.

Legislative history therefore offers tests for whether distinctions among an instrument, its events, its documents, its institutional bearers, and its scopes can be maintained without adopting contested normative conclusions. The formal profile catalog organizes these tests; it does not decide which legal theory of continuity is correct.

9.3 Cross-Institutional Provenance

Cross-institutional provenance highlights the distinction between a record and what the record purports to describe. A dataset entry, measurement report, transfer log, annotation, or derived metric is a candidate for REC. A copy may preserve content while creating a new token; an annotation or derivation may instead create a new version or informational object. Neither common content nor an unbroken provenance chain alone settles which record-identity criterion applies.

A sampled object, instrument, facility, or collection site is a candidate plain referent. The LOC profile asks about continuity of a specified site or position; the OBJ profile asks about continuity of a specified object. Forking record representations does not necessarily fork the represented object. Conversely, a process that separates, replaces, consumes, or transforms physical material may raise identity questions that preservation of a location or inventory label cannot resolve. The intended referent and transformation effects must be

established separately.

Producing organizations, custodians, laboratories, agencies, and recipients supply possible OBL carriers where their continuity and responsibility-bearing status are independently specified. Transfers, measurements, analyses, approvals, and custody changes supply possible OCC carriers. A transfer log is a record of a proposed transfer occurrence, not automatically that occurrence or proof of its identity.

Data-use policies, consent terms, retention rules, and access restrictions provide candidate rule referents for RULE-C or RULE-S. Institutional coverage conditions, consent domains, and data-sharing boundaries provide candidate scopes for SCOPE-E or SCOPE-S. Whether changes to policy text, permitted uses, scope coverage, or organizational structure preserve the relevant referent remains dependent on justified identity criteria.

These distinctions help separate shared reference from disagreements about data quality, classification, causation, responsibility, or permissible use. They also expose circumstances in which institutions lack enough warranted common reference for the Neutral Substrate assumption to be discharged. In that case, disputed co-reference or persistence judgments should not be treated as established merely because the records can be joined.

9.4 Summary

The three settings illustrate the analytic use of the selected carrier–basis catalog: OBL, OCC, and REC organize questions about bearers, occurrences, and records; LOC/OBJ, SCOPE-E/SCOPE-S, and RULE-C/RULE-S organize candidate comparisons between different bases for plain referents, scopes, and rules. Their usefulness lies in keeping these questions explicit, not in assuming that each proposed profile is a universally warranted regime.

Section 7 establishes the limits of the counting argument. The examples here neither supply a universal lower bound on irreducible regimes nor

independently validate the table's transformation classifications. To assess scientific adequacy, a concrete study must identify the intended referents and eligible presentations, justify its proposed identity criteria, and evaluate the resulting judgments against relevant examples and counterexamples.

Reference kinds select questions for analysis; declared bases propose criteria; transformation profiles state candidate consequences. Strict identity and persistence under transformation require further warrant, and the conditions for combining such criteria without loss remain open.

10 Related Work

This paper is situated at the intersection of formal ontology, provenance, and classification under disagreement. It does not propose a new upper ontology or derive a universally required inventory of identity regimes. Instead, it compares selected carrier–basis profiles under specified transformations and identifies what must still be established before those profiles can support warranted identity judgments.

10.1 Classification under Disagreement

Classification systems can embed institutional, social, and evaluative commitments rather than functioning as neutral descriptions (Bowker and Star 1999; Haslanger 2012). In scientific and policy settings, disagreement may persist because participants adopt different evidentiary, methodological, or evaluative frameworks, not merely because a temporary empirical uncertainty remains (Longino 1990). These observations motivate shared representations that do not presuppose interpretive convergence.

The present paper takes this motivation from *Neutral Substrates* (Case 2026a). That work establishes a conditional sufficiency result for semantic neutrality under Referential Common Ground; it does not supply independently justified

identity criteria for every intended referent. Here the narrower question is how proposed identity bases respond to specified record transformations, and what evidence would justify their use for shared reference. The selected profile catalog illustrates possible distinctions, not an irreducible minimum imposed by disagreement.

10.2 Identity, Roles, and Ontological Methodology

Formal ontology distinguishes identity criteria from ordinary classification and emphasizes the need to examine how roles and other properties relate to the identity of their bearers. OntoClean, in particular, analyzes rigidity and identity-related constraints on taxonomic organization (Guarino 1999; Guarino and Welty 2002). Accounts of social and institutional entities further examine bearers, roles, social objects, and institutional structures (Masolo et al. 2003; Guizzardi 2005).

This paper uses those distinctions as motivation, not as a derivation of its proposed identity bases. A role can be relevant to representing or tracking a referent without independently supplying a sufficient criterion for that referent's strict identity. Conversely, the use of an explicit role or basis parameter does not by itself make a representation inadequate. The question is which referent is intended, what conditions justify treating presentations as the same referent, and what a transformation actually preserves.

The paper compares declared transformation classifications under candidate identity bases. A difference between classifications is a formal distinction, but it does not alone prove a difference between generated equivalence relations or scientifically warranted identities.

10.3 Upper Ontologies

Upper ontologies such as BFO, DOLCE, and UFO provide resources for modeling entities, occurrences, dependence, qualities, roles, and institutional

or social objects (Smith et al. 2015; Arp et al. 2015; Gangemi et al. 2002; Guizzardi 2005). They address broader ontology-engineering purposes than the finite comparison undertaken here. This paper neither claims that those ontologies lack the necessary distinctions nor establishes that its selected profiles are required additions to them.

For example, a BFO-based model might represent a particular occurrence using an occurrent category and a material object using a continuant category, subject to the ontology's actual modeling commitments. Rules, records, and institutional bearers likewise require analysis of the intended referents and the resources available in a chosen ontology. Such mappings cannot be assumed solely from names in the selected inventory. They must be justified in the application concerned.

DOLCE- and UFO-based representations similarly offer resources for modeling social objects, roles, qualities, and dependence relations. A realization built with any of these frameworks can be examined for whether it distinguishes, where appropriate, a rule's content from its structure, a scope's coverage from its organization, a location from an occupying object, and a record from the thing recorded. Whether those are identity distinctions, rather than differences of description or representation, remains dependent on the selected referent and independently justified identity criterion.

Sections 5–7 compare such candidate distinctions and examine the limits of counting regimes. No result here demonstrates that an upper ontology must implement the selected profiles. The proposed comparison can instead help make modeling assumptions explicit and expose cases needing further scientific analysis.

10.4 Provenance, Versioning, and Exchange

The transformation families used here resemble operations encountered in provenance and versioning systems. PROV provides vocabulary for activities, entities, derivation, attribution, generation, use, and revision (Moreau and

Missier 2013). These relations may supply evidence about how representations were produced or changed. They do not, without additional identity conditions, settle whether two records are the same record, copies of a record, or records about the same referent.

An exchange formalism such as Common Logic may provide a setting for expressing some of the relevant commitments (ISO/IEC 2018). Conservative extension and heterogeneous ontology exchange are related logical concerns (Hodges 1997; Kutz et al. 2010). However, compatibility of a substrate with admissible frameworks is not, by itself, a proof of conservativity, and this paper constructs neither a Common Logic encoding nor a conservative-extension theorem. Its formal results concern the selected transformation profiles and the limits of conclusions drawn from them.

Related questions about long-lived reference and reuse arise in information infrastructures and digital-twin systems (Edwards 2010; Grieves and Vickers 2017). The candidate profiles offer questions to test in such settings: what is the intended bearer, which transformations preserve it, and what available evidence warrants those judgments? The paper does not establish a generally applicable solution for those systems.

10.5 Causal and Normative Separation

The distinction between observations or records and causal claims is important in causal analysis: causal conclusions depend on modeling assumptions beyond a descriptive record alone (Pearl 2009). The neutral-substrate constraint makes an analogous distinction at the foundational layer. A substrate may record that a specified source asserted a causal or normative proposition without itself adopting that proposition as fact. Under the assumptions of *Neutral Substrates*, the permitted foundational restriction is sufficient for semantic neutrality (Case 2026a).

The present paper examines a different issue: whether identity judgments about records, rules, occurrences, and their referents remain justified across

transformations. Annotation, copying, refinement, and provenance extension may preserve some features and change others. Which changes preserve a particular referent's strict identity is not determined merely by a transformation-family label. Keeping interpretation attributed also does not establish that the referential commitments themselves are adequately grounded.

10.6 FAIR Reuse under Persistent Disagreement

The FAIR principles motivate discoverable, accessible, interoperable, and reusable data across communities and over time (Wilkinson et al. 2016). They do not, by themselves, establish the referent-specific criteria needed to justify every co-reference or persistence judgment under persistent disagreement. Shared vocabularies, identifiers, schemas, and documented provenance can support reuse, but their presence alone does not prove strict identity.

The finite construction studied here complements these concerns by making selected identity-basis choices and proposed transformation effects explicit. Its nine candidate profiles can serve as comparison cases where different identity questions arise during reuse. Whether a particular case calls for separate profiles, an explicit carrier-indexed representation, or another adequate construction requires independent analysis. No particular irreducible-regime count follows from FAIR reuse or from the present catalog.

11 Limits and Boundary Conditions

The results of this paper concern a selected finite construction: six reference kinds, nine candidate carrier–basis profiles, and classifications over a stated transformation basis. Within that construction, the paper establishes specified classification-profile distinctions and the determinacy of a stipulated profile assignment.

11.1 Conditional Scope

The analysis is motivated by the neutral-substrate constraint of *Neutral Substrates* (Case 2026a) and by the setting of persistent interpretive disagreement described in Section 3. The neutrality result in that earlier work is conditional on Referential Common Ground and independently justified referential commitments. Neither the inventory selected here nor its transformation classifications establishes those premises for a concrete system.

The present results are relative to the selected carrier kinds, candidate identity bases, eligible transformation instances, and declared classification assignments. Their mathematical validity within the specified construction must be distinguished from the scientific adequacy of those choices. In particular, a record's stable label, a declared basis, or a non-breaking transformation classification does not, by itself, warrant strict identity across presentations.

The certified classification cells establish consequences of explicit invariant bases, effect models, and warrants. They do not establish that those models adequately describe the transformations or identity criteria of an actual system.

Individual derived cells may rely on different presentation spaces, field decompositions, and admissibility conditions. Their separate certification does not establish a coherent common semantic realization of the entire classification matrix.

Empirical counterexamples can challenge a proposed model or universal verdict. Finite observations alone do not establish universal conformance of an unrestricted implementation to that model.

The analysis is not tied to a particular legal theory, policy framework, or causal account. That independence of vocabulary does not prove that its selected identity conditions are valid in every such domain.

11.2 No Lower Bound, Upper Bound, or Completeness Claim

The size of the catalog in Section 6 is a property of the selected construction, not a proven minimum number of distinct identity relations or irreducible regimes required in every faithful realization. Nor does the catalog establish an upper bound on the possible identity bases or referent kinds.

Proposition 6.2 establishes that the declared classification functions are pairwise distinct. It does not establish that their generated equivalence relations are pairwise different on an eligible common domain, or that the respective identity judgments are scientifically warranted. The finite profile assignment of Theorem 8.5 is likewise deterministic within the selected catalog, not a derivation of which identity basis a real-world referent must have.

Proposition 7.2 establishes a narrower monotonicity statement: a classification difference persists under extension of the transformation basis when its earlier classifications are retained. It says nothing about the minimum number of irreducible regimes. Extending the basis may reveal additional candidate distinctions or inadequacies in previously assigned classifications; such outcomes require their own analysis.

The selected single-basis entries do not assert uniqueness. Records may require distinctions among content, version, token, signature, and chain of custody. Occurrences may be analyzed as bounded happenings, phases, or event complexes. Obligation-bearing parties may be distinguished from offices, roles, and organizational units. Whether any such distinction determines identity depends on an independently justified criterion for the intended referent.

Section 7 also gives a carrier-indexed combination of separate equivalence relations. That construction shows why distinct carrier kinds alone do not prove non-amalgamation. It does not establish that every identity account can be combined without loss. A minimum-regime theorem would require a justified unit of counting, adequate identity semantics, and a demonstrated

obstruction to all admissible combinations that purport to preserve the required judgments.

The carrier-indexed combination preserves its component equivalence relations, but does not establish that the resulting representation is scientifically adequate or preserves every required transformation distinction.

11.3 Limits of the Selected Setting

The following cases expose questions that the selected construction does not settle. They should not be read as blanket exclusions of otherwise legitimate representational methods.

Interpretively dependent identity. A framework may supply a causal, normative, or institutional criterion under which a referent is identified. Such a judgment can be recorded as an attributed interpretation. Treating it as an unqualified shared referential commitment requires independent grounds and compatibility with the applicable admissible frameworks. The present profile catalog does not establish those grounds.

Relation-supported identity. Relations may provide evidence or conditions for individuation, co-reference, or persistence. Their role cannot be ruled out by stipulation. When a construction uses relational criteria, it must explain what they establish, for which referents, and under what interpretation conditions. This paper does not derive a general prohibition on relation-determined identity.

Framework-relative identity judgments. Admissible frameworks may disagree about co-reference or persistence. Recording their distinct judgments does not itself establish a common identity judgment. If the substrate adopts a contested judgment as foundational, the Referential Common Ground

premise must be re-examined. Disagreement alone, however, does not prove that all shared reference has failed.

Declared parameters and undeclared distinctions. A role, flag, carrier tag, or identity-basis parameter is not intrinsically a hidden regime. An explicit parameter can preserve a meaningful distinction within a combined representation. Conversely, an undocumented discriminator or a parameter whose meaning varies across participating frameworks may fail to support warranted shared identity judgments. Whether a proposed combination is faithful must be assessed rather than decided by its number of labels.

Unmodeled transformations. The selected transformation basis $\mathcal{F}$ is not complete. An application may involve operations whose effects differ from the proposed family-level assignments or require finer typing. Such cases limit the adequacy of the current catalog for that application.

11.4 Scope Exclusions

This paper does not construct causal or normative interpretive frameworks, determine which interpretation should govern, or prescribe a universal ontology. It does not provide a method for discovering the correct identity criterion for an arbitrary referent, certify the soundness of transformation classifications in deployed systems, or establish a general procedure for deciding strict identity.

The paper also does not prescribe that every identity-relevant distinction must be implemented as a separate regime object. A construction using explicit carrier or basis parameters may be suitable if its semantics are fixed, its judgments are warranted, and the required distinctions are preserved. A context-relative construction may serve a different objective, provided the limits on shared reference and interpretation are stated. The scientific questions are what the construction preserves and why those preservation

claims are justified, not merely where a distinction appears in a schema.

11.5 Boundary with Accountable Records

This paper investigates candidate identity bases and their proposed behavior under transformations of representations. It does not provide a conformance checker for deployed record systems, mandate a schema for basis declarations or transformation logs, or establish that an implementation discloses every identity-relevant mechanism and use.

Related operational-identity and accountable-records work can examine whether implemented behavior agrees with a declared criterion and whether relevant evidence is observable. Such an audit cannot, solely from agreement with a declaration, certify that the declaration supplies a scientifically adequate identity criterion. Conversely, the finite construction here does not impose a minimum number of necessary implementation categories on such an audit.

11.6 Boundary of the Claim

Within the specified finite construction, the named profiles are pairwise distinct as classification functions, the selected carrier–basis assignment is determinate, and existing classification distinctions persist under extensions that retain the original classifications. The conditional transformation comparisons show how different proposed identity bases can assign different effects to an operation family.

These are limited formal results. They do not establish that distinct classification profiles yield distinct scientifically warranted identity relations or that every neutral accountability substrate must realize a minimum number of irreducible regimes. Establishing such a necessity claim would require additional, independently justified conditions; the present paper leaves that question open.

12 Conclusion

Naming a reference kind does not settle the identity of its referents. Nor does classifying a transformation as preserving or breaking identity, by itself, establish a warranted judgment of sameness. Such judgments require independently justified individuation, co-reference, and persistence criteria for the intended referents. Transformation analysis can make proposed criteria and their consequences explicit, but it cannot supply their scientific justification by stipulation.

This paper takes its motivation from the conditional neutrality result in *Neutral Substrates* (Case 2026a). Under Referential Common Ground and the stated restriction on foundational commitments, that result provides sufficient conditions for semantic neutrality. It does not establish the identity criteria required for particular referents. The present paper examines selected candidate criteria and how they classify transformations of representations.

The selected reference inventory and candidate carrier–basis profiles are presented in Section 6.

Three paired comparisons motivate closer investigation: coverage versus internal structure for scopes, content versus structure for rules, and locus versus object for plain referents. Decomposition, refinement, and forking supply conditional tests of these proposed distinctions, not universal identity laws.

Within the selected finite construction, the declared classification functions are pairwise distinct, the specified carrier–basis assignment is determinate, and classification differences persist when the transformation basis is extended without changing existing assignments. These results concern declared profiles and their formal properties. A difference between classifications does not necessarily produce a difference between generated equivalence relations, and neither establishes strict identity or non-identity without further warrant.

The carrier-indexed combination considered in Section 7 shows why dis-

tinguishable relations need not be counted as separate formal structures. Whether a proposed combination faithfully preserves all warranted identity judgments and transformation distinctions remains a substantive question, as does any proposed non-amalgamation condition.

The remaining work is therefore both mathematical and scientific: justify the proposed identity bases for their intended referents; validate transformation classifications on eligible presentations; distinguish generated relations from warranted co-reference and persistence judgments; and determine what, if anything, requires irreducibly separate representations. The application settings in Section 9 offer contexts in which to test these conditions, rather than evidence that the selected catalog is complete.

The contribution of this paper is a conditional formal method for deriving and auditing identity-preservation classifications under transformation.

The certified model-derived cells demonstrate how explicitly declared identity invariants, transformation effects, and additional warrants determine classification verdicts. The step-generated relation results establish precise conditions for equality and separation of the resulting formal identity relations.

These results make proposed identity claims more auditable, provide testable conditions for transforming systems, and identify assumptions that must be independently justified.

The remaining scientific questions concern the adequacy of the identity criteria, the correspondence between formal models and real transformations, and the compatibility of the separately modeled classifications.

Companion operational-identity and accountable-records work can examine agreement between implementations and declared criteria; that agreement alone cannot certify the criteria's scientific adequacy.

Stable reference under persistent disagreement remains the objective.

Statements and Declarations

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Author Contributions

The author is the sole contributor to this work and is responsible for all aspects of the research, authorship, and publication.

Use of AI-Assisted Tools

AI-assisted tools were used for editing, formatting, and consistency checking, and in developing and reviewing mathematical definitions, proofs, and the companion Lean 4 formalization. The author reviewed all suggestions and is solely responsible for the content.

Declaration of Conflicting Interest

The author declares no potential conflicts of interest.

A Pairwise Distinctions and Their Limits

The nine candidate profiles in Table 1 yield $\binom{9}{2} = 36$ unordered pairs. The declared classification functions are different for each pair, as established by Proposition 6.2. This is a result about the *proposed matrix*, not a proof that all pairs induce distinct identity relations or represent scientifically necessary identity criteria.

A.1 Family-Level Generated Relations

Definition 3.5 generates $\sim_\tau$ using admissible steps of families classified PRS or NEU.

This construction selects steps by their stipulated family classifications rather than independently testing invariant preservation at the individual-step level.

Within the stipulated matrix, OBL and REC have identical sets of non-breaking families. Consequently, on a common eligible domain with the same admissible transformation dynamics, this paper's construction generates equal relations for those profiles. This is a formal comparison; the distinct intended carrier kinds have not been shown to share a scientifically warranted comparison domain.

The companion Persistence classification construction differs: it generates relations using only families licensed PRS.

Within the stipulated matrix, OBL/RULE-C and SCOPE-S/RULE-S have identical sets of PRS families. They therefore generate identical relations under that construction on shared transformation dynamics.

Direct enumeration of the stipulated matrix gives seven distinct PRS-only family sets and eight distinct PRS-or-NEU family sets.

These counts bound the number of distinct relations generated on common dynamics by the respective classification-based constructions. They do not

count scientifically required regimes and do not bound the invariant-based step relations in general.

The remaining 34 unordered pairs have different sets of PRS families and can be separated under suitable dynamics. Such differences do not prove separation on arbitrary dynamics, which may lack steps witnessing those differences.

These PRS-only results do not establish corresponding equality or separation results for this paper's PRS-or-NEU generated relation.

A.2 Invariant-Generated Relations

The newer invariant-based construction includes the admissible individual steps that preserve a declared invariant, rather than selecting entire families solely by their matrix labels.

Theorem 6.4 establishes that two bases on a common effect model and comparison domain generate equal relations exactly when they agree on invariant preservation for every admissible step in that domain.

Proposition 6.5 establishes that an admissible step preserved by one basis and changed by another is a sufficient witness of different generated relations, provided both endpoints lie in the comparison domain.

Conversely, bases that classify a family differently can still produce equal relations on a restricted domain containing no separating step.

None of these three generated-relation constructions establishes strict identity without an independently justified interpretation.

A.3 Conditional Examples

The scope, rule, and fork models discussed in Section 6.1 demonstrate how explicit presentation fields, transformation effects, and warrants determine

derived verdicts. The fork model is especially instructive: a bare fork does not entail either a change of locus or a change of object. The proposed LOC/OBJ distinction therefore requires additional continuity warrants. Whether the resulting generated relations coincide or differ also depends on the admissible steps and comparison domain.

The stipulated matrix establishes pairwise differences among the proposed classification profiles. These differences do not necessarily imply distinct generated relations. The invariant-based step results provide explicit conditions for relation equality and separation, while their interpretation as strict identity requires independent scientific justification.

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